

Table 13-2. Comparison Between the Errors of $\pi(x)$ and $\text{Li}(x)$ Approximations and the Deterministic Radial Formula Developed in This Study

10^n	$\pi(10^n)$ (actual prime counts)	C_cand	C_prime (= $\pi(10^n) - 4$)	C_comp	Li(x)	Li- π
10^1	4	0	0	0	5.1	1.1
10^2	25	24	21 (= 25 - 4)	3	29.1	4.1
10^3	168	264	164 (= 168 - 4)	100	176.6	8.6
10^4	1,229	2,664	1,225 (= 1,229 - 4)	1,439	1245.1	16.1
10^5	9,592	26,664	9,588 (= 9,592 - 4)	17,076	9628.8	36.8
10^6	78,498	266,664	78,494 (= 78,498 - 4)	188,170	78626.5	128.5
10^7	664,579	2,666,664	664,575 (= 664,579 - 4)	2,002,089	664917.4	338.4
10^8	5,761,455	26,666,664	5,761,451 (= 5,761,455 - 4)	20,905,213	5762208.3	753.3
10^9	50,847,534	266,666,664	50,847,530 (= 50,847,534 - 4)	215,819,134	50849233.9	1699.9
10^{10}	455,052,511	2,666,666,664	455,052,507 (= 455,052,511 - 4)	2,211,614,157	455055613.5	3102.5
10^{11}	4,118,054,813	26,666,666,664	4,118,054,809 (= 4,118,054,813 - 4)	22,548,611,855	4118066399.6	11586.6
10^{12}	37,607,912,018	266,666,666,664	37,607,912,014 (= 37,607,912,018 - 4)	229,058,754,650	37607950279.8	38261.8
10^{13}	346,065,536,839	2,666,666,666,664	346,065,536,835 (= 346,065,536,839 - 4)	2,320,601,129,829	346065645809.0	108970.0
10^{14}	3,204,941,750,802	26,666,666,666,664	3,204,941,750,798 (= 3,204,941,750,802 - 4)	23,461,724,915,866	3204942065690.9	314888.9
10^{15}	29,844,570,422,669	266,666,666,666,664	29,844,570,422,665 (= 29,844,570,422,669 - 4)	236,822,096,243,999	29844571475286.5	1052617.5
10^{16}	279,238,341,033,925	2,666,666,666,666,664	279,238,341,033,921 (= 279,238,341,033,925 - 4)	2,387,428,325,632,743	279238344248555.8	3214630.8
10^{17}	2,623,557,157,654,233	26,666,666,666,666,664	2,623,557,157,654,229 (= 2,623,557,157,654,233 - 4)	24,043,109,509,012,435	2623557165610820.5	7956587.7
10^{18}	24,739,954,287,740,860	266,666,666,666,666,664	24,739,954,287,740,856 (= 24,739,954,287,740,860 - 4)	241,926,712,378,925,808	24739954309690412.0	21949554.0
10^{19}	234,057,667,276,344,607	2,666,666,666,666,666,664	234,057,667,276,344,603 (= 234,057,667,276,344,607 - 4)	2,432,608,999,390,322,061	234057667376222368.0	99877774.2
10^{20}	2,220,819,602,560,918,840	26,666,666,666,666,666,664	2,220,819,602,560,918,836 (= 2,220,819,602,560,918,840 - 4)	24,445,847,064,105,747,828	2220819602783663360.0	222744642.5
10^{21}	21,127,269,486,018,731,928	266,666,666,666,666,666,664	21,127,269,486,018,731,924 (= 21,127,269,486,018,731,928 - 4)	245,539,397,180,647,934,740	21127269486616125440.0	597394253.3
10^{22}	201,467,286,689,315,906,290	2,666,666,666,666,666,666,664	201,467,286,689,315,906,286 (= 201,467,286,689,315,906,290 - 4)	2,465,199,379,977,350,760,378	201467286691248275456.0	1932355207.1
10^{23}	1,925,320,391,606,803,968,923	26,666,666,666,666,666,666,664	1,925,320,391,606,803,968,919 (= 1,925,320,391,606,803,968,923 - 4)	24,741,346,275,059,862,697,745	1925320391614054006784.0	7250186214.7
10^{24}	18,435,599,767,349,200,867,866	266,666,666,666,666,666,666,664	18,435,599,767,349,200,867,862 (= 18,435,599,767,349,200,867,866 - 4)	248,231,066,899,317,465,798,802	18435599767366346473472.0	17146907277.1
10^{25}	176,846,309,399,143,769,411,680	2,666,666,666,666,666,666,666,664	176,846,309,399,143,769,411,676 (= 176,846,309,399,143,769,411,680 - 4)	2,489,820,357,267,522,897,254,988	176846309399198921916416.0	55160980938.3